higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL SURVEYING N6

3 APRIL 2019

This marking guideline consists of 7 pages.

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QUESTION 1

NAME	JOIN	ΔY	ΔX	NAME	Y	X
P				P	+458,82	+1 532,32
232:39:10		-197,19√	-150,47√			
248,04 m		-0,20√	+0,13√			
R1				R1	+261,43✓	+1 381,98✓
286:56:08		-258,49√	+78,71√			
270,21 m		-0,22√	+0,15√			
R2				R2	+2,72✓	+1 460,84✓
327:11:36		-115,75√	+179,56√			
213,63 m		-0,17√	+0,12√			
R3				R3	-113,20✓	+1 640,52✓
38:41:50		+74,51√	+93,00√			
119,17 m		-0,09√	+0,06√			
Q				Q	-38,78	+1 733,58
851,05✓		-496,92√	+200,80√		-497,60√	+201,26√
		-497,60√	+201,26√			
		-0,68√	+0,46√			

$\frac{-0,68}{851,05}$ X leg√ $\frac{+0,46}{851,05}$ X leg√ **[20]**

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QUESTION 2

Station		Distance		HI or middle hair MH	Angles		HI – MH + -	Height component + -	Height difference + -	Elevation of point	Remarks
From	To	Stadia	Hor		Hor	Vert					
T				1,46						713,24	
	T1	1,90 0,58	129,05√	1,24✓	136:49:00	98:36:00	+0,22✓	-19,96√	-19,74✓	693,50✓	
	T2	4,02 2,60	139,01√	3,31✓	173:21:00	81:39:00	-1,85✓	+20,40√	+18,55✓	731,79✓	
	T3	1,84 0,60	121,69√	1,22✓	246:32:00	82:09:00	+0,24✓	+16,78√	+17,02✓	730,26✓	
	T4	3,64 1,26	228,38√	2,45✓	299:52:00	101:36:00	-0,99✓	-46,88√	-47,87✓	665,37✓	

1 × 16 = 16
0,5 × 8 = 04
= 20 **[20]**

QUESTION 3

3.1 A peg/marker in the ground used as a boundary peg, control point, benchmark, etc.

3.2 A line which lies in the level surface and is therefore normal to the direction of gravity at all times

3.3 Lines on the surface of the earth joining points of the same height above mean sea level

3.4 Entering of measurements in a systematic, logical manner so that any draughtsmen can produce a map or a plan from the recorded information

3.5 Placing of pegs in the ground to define the position of pegs/beacons/survey stations, mark out the limits of a structure and refer to a vertical height

(5 × 2) **[10]**

QUESTION 4

4.1 Join AB:

Dy_{AB} = +4 328,66 – (+4 896,26)
= -567,60 m✓

Dx_{AB} = +3 844,24 – (+3 926,82)
= -82,58 m✓

Direction AB = $\frac{-567,60}{-82,58}$ 3rd quadrant

Tanθ = 261:43:19✓

Direction AB 261:43:19

Angle BAC 26:25:31

AC 288:08:50✓

Angle AB 261:43:19
BA 81:43:19✓
Angle ABC -29:53:47
BC 51:49:32✓ (6)

(8)

4.2

A			+4 896,26	+3 926,82
288:08:50	343,55Sin 288°08'50"	343,55Cos 288°08'50"	-326,46	+107,00
343,55	= -326,46✓✓	= +107,00✓✓		
C			+4 569,80✓	+4 033,82✓
A	+4 896, 26	+3 926,82		
C	+4 569,80	+4 033,82		
Checked	-326,46✓	+107,00✓		

[14]

QUESTION 5

	Height	Times used	Product
A1	139,44	1	139,44√
A2	136,88	2	273,76√
A3	128,46	1	128,46√
B1	152,46	2	304,92√
B2	138,78	4	555,12√
B3	129,34	2	258,68√
C1	160,55	1	160,55√
C2	156,68	2	313,36√
C3	143,86	1	143,86√
		16	2 278,15√

Mean height = $\frac{2\ 278,15}{16}$

= 142,38 m✓

= 142,38 – 125,00

= 17,38 m✓

V = [4(25 × 25)] × 17,38

= 43 450 m^3✓

15% = 43 450 × 0,15

= 6 517,50 m^3✓

Total volume = 43 450 + 6 517,50

= 49 967,50 m^3✓ **[10]**

QUESTION 6

$$T = R\tan\frac{\Delta}{2}$$

$$T = 218,50 \times \tan 19:40:30 \checkmark$$

= 78,13 m → ✓

BC = IP – T✓

BC = 3 372,33 – 78,13✓

BC = 3 294,20 m✓

$$La = \frac{\pi \Delta R}{180}$$

$$= \frac{\pi \times 39:21:00 \times 218,5}{180} \checkmark$$

La = 150,06 m✓

EC = BC + La✓

= 3 294,20 + 150,06✓

EC = 3 444,26 m✓

(a = 5,8) $= \frac{1\ 718,9 \times 5,8}{218,5}$ = 00:45:37✓

(a = 20) $= \frac{1\ 718,9 \times 20}{218,5}$ = 02:37:20✓

(a = 4,26) $= \frac{1\ 718,9 \times 4,26}{218,5}$ = 00:33:30✓

Chainage	Chord	ϑ	α
BC 3 294,20		00:00:00	360:00:00√
3 300,00	5,80	00:45:37	359:14:23√
3 320,00	20,00	02:37:20	356:37:03√
3 340,00	20,00	02:37:20	353:59:43√
3 360,00	20,00	02:37:20	351:22:23√
3 380,00	20,00	02:37:20	348:45:03√
3 400,00	20,00	02:37:20	346:07:43√
3 420,00	20,00	02:37:20	343:30:23√
3 440,00	20,00	02:37:20	340:53:03√
EC 3 444,26	4,26	00:33:30	340:19:33√
150,06√	150,06√	19:40:27√	19:40:27√

1 × 13 = 13
0,5 × 14 = 7
20 **[20]**

QUESTION 7

At A:

Station	Circle left	Circle right	Angle circle left	Angle circle right	Mean angle
B2	129:44:50	230:13:48	39:44:50✓	39:46:12✓	-39:45:31✓
B5	81:08:24	278:52:18	08:51:36✓	08:52:18✓	+08:51:59✓

[6]

TOTAL: 100

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